



Isothermal v.l.e. and excess molar Gibbs energy of binary and ternary mixtures containing diisopropyl ether, *n*-heptane and isopropanol at $T = 313.15$ K

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(Vapour + liquid) equilibrium of {diisopropyl ether (DIPE) + *n*-heptane + isopropanol}, {diisopropyl ether (DIPE) + *n*-heptane}, {diisopropyl ether (DIPE) + isopropanol} and (*n*-heptane + isopropanol) were measured at $T = 313.15$ K. Data reduction by Barker's method provides correlations for the excess molar Gibbs energy G_m^E using the Margules equation for the binary systems and the Wohl expansion for the ternary. The Wilson, NRTL, and UNIQUAC models were applied successfully to both the binary and the ternary systems reported here. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Diisopropyl ether (DIPE) is one of the new oxygenated additives which has potential use as a blending agent in the formulation of unleaded gasoline because of its antiknocking effect and its environment-friendly nature.

DIPE is obtained by the reaction of isopropanol and propene. Isopropanol itself is synthesised by hydrolysis of propene, hence DIPE production is linked to the availability of the latter. We have chosen *n*-heptane as the third compound to represent the hydrocarbons in the gasoline.

This work is part of a research programme on the thermodynamic characterization of ternary mixtures containing oxygenated additives (ethers and alcohols) and different types of hydrocarbons (paraffins, cycloparaffins, aromatics, olefins). In previous work,^(1–8) we have investigated binary and ternary systems containing MTBE (methyl *tert*-butyl ether),

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benzene, cyclohexane, *n*-heptane, 1-hexene, and methanol at $T = 313.15$ K. In addition, the oxygenated additive TAME (*tert*-amyl ether) with the same compounds has been investigated and the results have been published.^(9,10)

The (vapour + liquid) equilibria of (DIPE + *n*-heptane + isopropanol), (DIPE + *n*-heptane), (DIPE + isopropanol), and (isopropanol + *n*-heptane) at $T = 313.15$ K form the object of the present paper. A similar investigation on (DIPE + cyclohexane + benzene) and its constituent binaries has been recently reported.^(11,12)

2. Experimental

All the chemicals used were purchased from Fluka Chemie AG and were of the highest purity available, chromatography quality reagents (of the series puriss. p.a.) with a mole fraction purity >0.99 for DIPE and >0.995 for *n*-heptane and isopropanol. All liquids were degassed prior to use using a modified distillation method based on the technique of Van Ness and Abbott,⁽¹³⁾ under a vacuum generated by a double stage rotatory pump assuring 0.5 Pa. The mole fractions purities of the chemicals were checked by gas chromatography and were found to be: >0.995 for diisopropyl ether and >0.998 for *n*-heptane and isopropanol.

A static (vapour + liquid) equilibrium (v.l.e.) apparatus consisting of an isothermal total pressure cell^(1,8) was employed for measuring the (vapour + liquid) equilibrium of the binary and ternary mixtures. The technique was developed by Van Ness and coworkers.^(14,15)

Three positive displacement pumps of 100 ml capacity (Ruska, mod. 2200–801) equipped with piston injectors were used to inject known volumes of degassed components into a cell immersed in a high precision water bath (Hart Scientific model 6020) assuring a temperature stability of ± 0.5 mK and thermostatted at $T = 313.15$ K. The pump resolution is 0.01 ml and the resulting uncertainty in the volume injected is ± 0.03 ml.

The cell is a cylindrical stainless steel vessel with a capacity of about 180 ml and is provided with an externally-operated magnetic stirrer. Initially about 50 ml of one component are injected into the evacuated cell and the vapour pressure is recorded. The second and third components are then injected in appropriate proportions so as to achieve a desired composition. The total mass injected is determined from the volume differences corresponding to the initial and final positions of the pistons, the temperature of the injectors and the densities of the injected components, allowing an uncertainty in the mole fraction of less than $\pm 5 \cdot 10^{-4}$, without the need to sample the phases.

The total vapour pressure for a ternary mixture is obtained by adding a third component to a binary mixture at a fixed temperature. Six runs (dilution lines) were carried out starting with a binary mixture with a mole fraction of one component close to $x = (0.3, \text{ or } 0.7)$, and adding the third component up to a mole fraction $x = 0.5$.

The temperature was measured by a calibrated standard PRT-100 (SDL model 5385/100) connected to an a.c. resistance bridge (ASL model F250) with a temperature resolution of 1 mK. The estimated uncertainty of the temperature measurement is ± 10 mK. The pressure was measured using a differential pressure cell provided with a null indicator (Ruska models 2413–705 and 2416–711, respectively). When atmospheric air balances

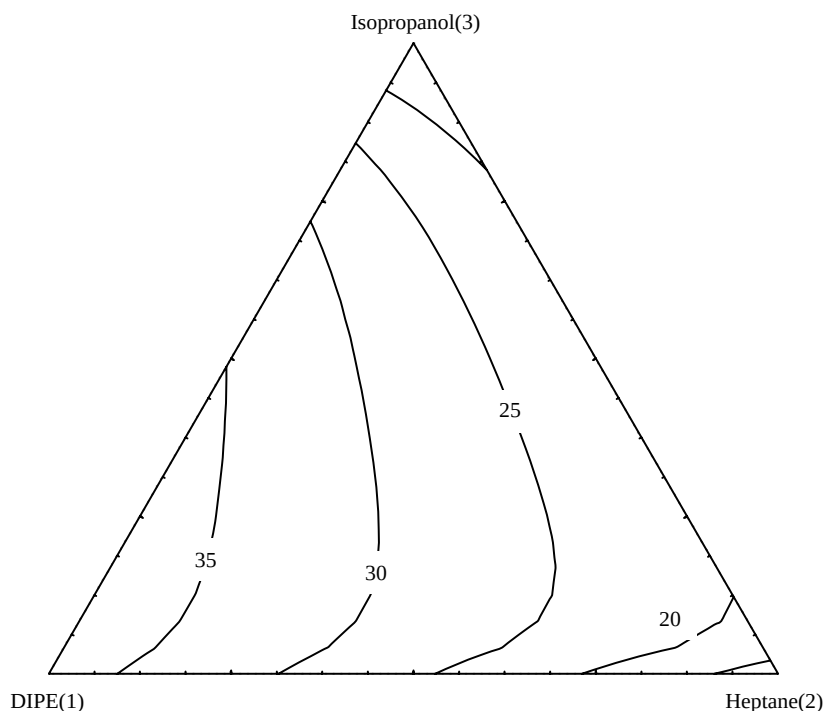


FIGURE 1. Lines of constant total pressure p using the Wohl expansion model for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K.

the vapour pressure of the cell a Bourdon fused quartz precision pressure gauge (Texas Instruments model 801) provided with a capsule indicates the pressure with an estimated uncertainty of ± 5 Pa for the 125 kPa range. Both temperature and pressure devices have been calibrated using our own standards which are traceable to S.I. units.

3. Results and correlations

The use of the measuring technique described above allows a static equilibrium between the phases present, assuring a true thermodynamic equilibrium. Direct sampling, particularly of the vapour phase, upsets the equilibrium, the mass of the vapour in the cell being very small, yet an appreciable mass must be withdrawn to yield an amount of condensate sufficient for accurate analysis. However, as a consequence of Duhem's theorem, sampling of the phases is, in fact, not necessary. Given a set of equilibrium (x, p) data at constant T , thermodynamic relations allow the calculation of the y values. Thus, the vapour phase needs not to be sampled and the resulting data are thermodynamically consistent *per se*.^(16–18)

The data reduction of the binary and ternary mixtures has been performed using Barker's method⁽¹⁹⁾ according to well established procedures.^(20,21)

TABLE 1. Values used for the reduction of the data: experimental vapour pressures p_i^S , molar volumes V_i^L , and second virial coefficients B_{ii} and B_{ij} of the pure compounds used in this work, and corresponding literature values at $T = 313.15$ K

	$((\text{CH}_3)_2\text{CH})_2\text{O}(1)$	$\text{CH}_3(\text{CH}_2)_5\text{CH}_3(2)$	$(\text{CH}_3)_2\text{CHOH}(3)$
p_i^S/kPa	37.125	12.356	13.903
	37.081 ^a	12.300 ^c	13.853 ^a
	37.090 ^b	12.335 ^d	13.864 ^g
		12.348 ^e	13.872 ^g
		12.331 ^f	13.895 ^h
			13.910 ⁱ
$V_i^L/(\text{cm}^3 \cdot \text{mol}^{-1})$	145.4 ^j	150.3 ^j	78.2 ^j
$B_{11}/(\text{cm}^3 \cdot \text{mol}^{-1})$	-1687.8 ^k	-2025.4 ^k	-1381.6 ^k
$B_{12}/(\text{cm}^3 \cdot \text{mol}^{-1})$	-2025.4 ^k	-2520.5 ^k	-1166.6 ^k
$B_{13}/(\text{cm}^3 \cdot \text{mol}^{-1})$	-1381.6 ^k	-1166.6 ^k	-1878.3 ^k

^a Calculated from Antoine equation using constants reported by Riddick *et al.*⁽²⁹⁾; ^b Calculated from Antoine equation using constants reported by Ambrose *et al.*⁽³⁰⁾; ^c Calculated from Antoine equation using constants reported by Reid *et al.*⁽³¹⁾; ^d Calculated from Antoine equation using constants reported in TRC⁽³²⁾; ^e Reported by Montero⁽³⁴⁾; ^f Reported by Segovia⁽⁴⁾; ^g Reported by Mara *et al.*⁽³⁵⁾; ^h Reported by Ambrose *et al.*⁽³⁶⁾; ⁱ Reported by Oracz⁽³⁷⁾; ^j Reported in TRC^(32,33); ^k Calculated by Hayden *et al.*⁽¹⁹⁾ from Dymond *et al.*⁽²⁰⁾

The non-ideality of the vapour phase is taken into account by the virial equation of state, truncated at the second term. The second virial coefficients have been calculated by the Hayden O'Connell method⁽²²⁾ using the coefficients given by Dymond *et al.*⁽²³⁾

The ternary $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ and corresponding binaries $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)\text{CH}_3(\text{CH}_2)_5\text{CH}_3]$; $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)(\text{CH}_3)_2\text{CHOH}]$ and $[x_1(\text{CH}_3)_2\text{CHOH} + (1 - x_1)\text{CH}_3(\text{CH}_2)_5\text{CH}_3]$ have been investigated at $T = 313.15$ K. The data of the ternary mixture are adequately correlated by the three-parameter Wohl expansion,⁽²⁴⁾ equation (1):

$$g_{123} = G_m^E/RT = g_{12} + g_{13} + g_{23} + (C_0 + C_1x_1 + C_2x_2)x_1x_2x_3, \quad (1)$$

which includes also the parameters of the corresponding binaries. The adjustable parameters C_0 , C_1 and C_2 are found by regression of the ternary data.

Correlations for the g_{ij} are given by the following six parameter Margules equation⁽²⁵⁾

$$g_{ij} = G_m^E/RT = \{A_{ji}x_i + A_{ij}x_j - (\lambda_{ji}x_i + \lambda_{ij}x_j)x_ix_j + (\eta_{ji}x_i + \eta_{ij}x_j)x_i^2x_j^2\}x_ix_j. \quad (2)$$

The binary and ternary systems have also been correlated using the Wilson,⁽²⁶⁾ NRTL,⁽²⁷⁾ and UNIQUAC⁽²⁸⁾ models, whose respective excess Gibbs energy expressions are given

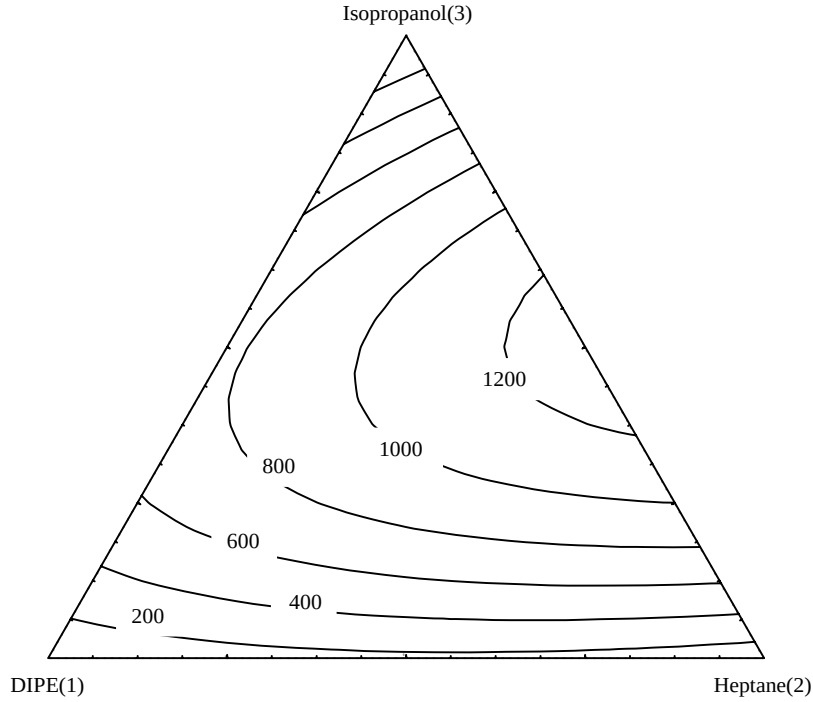


FIGURE 2. Lines of constant excess molar Gibbs energy G_m^E using the Wohl expansion model for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K.

by the following equations:

$$G_m^E/RT = -\sum_i x_i \ln\left(\sum_j x_j A_{ij}\right), \quad (3)$$

$$G_m^E/RT = \sum_i x_i \left(\sum_j A_{ji} G_{ji} x_j / \sum_k G_{ki} x_k \right), \quad (4)$$

$$G_m^E/RT = \sum_i x_i \ln(\varphi_i/x_i) + z/2 \sum_i q_i x_i \ln(\vartheta_i/q_i) - \sum_i q_i x_i \ln\left(\sum_j \vartheta_j A_{ji}\right), \quad (5)$$

where $G_{ji} = \exp(-\alpha_{ji} A_{ji})$, $\vartheta_i = q_i x_i / \sum_j q_j x_j$, $\varphi_i = r_i x_i / \sum_j r_j x_j$, and $z = 10$.

The average values of the experimental vapour pressures (p_i^s) of the pure components compared with literature values, their molar liquid volumes (V_i^L) and second virial coefficients (B_{ii} , B_{ij}) used in the calculations are presented in table 1.

Tables 2 to 5 give the experimental values of the total pressure p and the corresponding compositions of the liquid x_i and vapour phases y_i as reduced by the Margules equation for the binary mixtures and the Wohl expansion for the ternary ones.

TABLE 2. Total pressure p for $[x\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1-x)\text{CH}_3(\text{CH}_2)_5\text{CH}_3]$ at $T = 313.15$ K, and at various compositions of the liquid x and vapour phases y

x	y	p/kPa	x	y	p/kPa
0.0000	0.0000	12.345	0.4991	0.7462	25.617
0.0534	0.1602	13.931	0.5493	0.7798	26.806
0.1036	0.2779	15.399	0.5494	0.7799	26.815
0.1500	0.3668	16.695	0.5990	0.8104	27.965
0.1999	0.4468	18.074	0.5991	0.8104	27.985
0.2536	0.5194	19.497	0.6500	0.8395	29.160
0.3000	0.5734	20.705	0.6990	0.8655	30.273
0.3512	0.6257	22.002	0.7528	0.8924	31.522
0.3994	0.6693	23.203	0.8039	0.9166	32.671
0.4010	0.6706	23.239	0.8584	0.9410	33.913
0.4494	0.7098	24.423	0.9060	0.9615	34.996
0.4499	0.7101	24.433	0.9581	0.9831	36.161
0.4989	0.7461	25.612	1.0000	1.0000	37.134

TABLE 3. Total pressure p for $[x\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1-x)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K, and at various compositions of the liquid x and vapour phases y

x	y	p/kPa	x	y	p/kPa
0.0000	0.0000	13.902	0.4995	0.7378	34.826
0.0537	0.2865	18.558	0.5487	0.7544	35.533
0.1037	0.4258	22.027	0.5497	0.7547	35.543
0.1448	0.4999	24.422	0.5998	0.7708	36.160
0.2000	0.5689	27.074	0.6000	0.7709	36.160
0.2502	0.6139	29.005	0.6489	0.7865	36.687
0.2972	0.6466	30.528	0.7002	0.8033	37.152
0.3497	0.6760	31.921	0.7467	0.8197	37.509
0.3994	0.6992	33.045	0.7979	0.8399	37.822
0.3995	0.6993	33.062	0.8516	0.8656	38.025
0.4491	0.7194	33.998	0.9007	0.8958	38.051
0.4495	0.7196	34.016	0.9535	0.9409	37.784
0.4983	0.7374	34.818	1.0000	1.0000	37.127

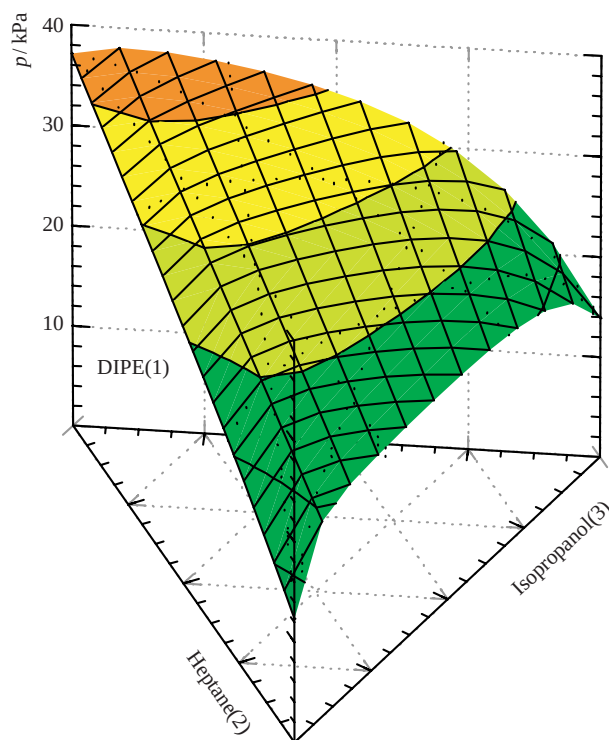


FIGURE 3. Pressure surface p using the Wohl expansion model for $[x_1(\text{CH}_3)_2\text{CH}]_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K.

The data correlation results for the binary systems reported here are summarized in table 6. This table contains the values of the adjustable parameters of the various models which lead to the best results using Barker's method, the root-mean-square deviation (r.m.s.d.) of the difference between the experimental and the calculated pressure and the maximum value of this difference ($\max |\Delta p|$). For the ternary system, the results of the correlation are given in table 7.

The results for the ternary system are shown in figures 1 to 4. Constant pressure lines are shown in figure 1, constant excess molar Gibbs energy lines G_m^E in figure 2 and pressure and excess molar Gibbs energy surfaces in figures 3 and 4, respectively, using the Wohl expansion model.

As an example of graphic representation of binary systems we have chosen $[x_1(\text{CH}_3)_2\text{CH}]_2\text{O} + (1 - x_1)(\text{CH}_3)_2\text{CHOH}]$, the most non-ideal system showing azeotropy of the three will be discussed later. Figure 5 represents the experimental pressure against the composition of the liquid and vapour phases, and figure 6 the pressure residuals, defined as differences between the experimental and calculated pressures using the four-parameter Margules equation, as a function of the DIPE liquid phase composition.

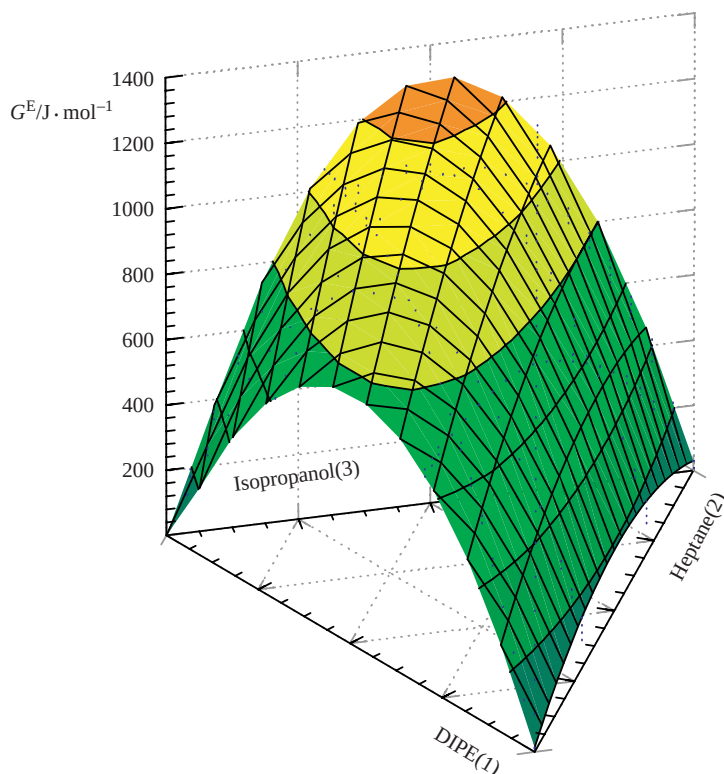


FIGURE 4. Excess molar Gibbs energy G_m^E using the Wohl expansion model for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K.

In this latter figure are also inserted dashed lines corresponding to ± 0.1 per cent deviation of the experimental pressure of each measured point, as an illustration of the quality of the experimental data.

4. Discussion

The experimental data for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ have been correlated using the Wohl equation which is, in fact, an extension of the Margules equation to multicomponent systems, the Wilson, NRTL, and UNIQUAC models. The minimum value of the root mean square deviation (r.m.s.d.) of Δp , obtained with NRTL model, is 25 Pa and the maximum difference between the calculated and experimental pressure ($\max|\Delta p|$) is 77 Pa. The pressure surface, figure 3, shows a maximum at 38.037 kPa which corresponds to the azeotrope $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)(\text{CH}_3)_2\text{CHOH}]$, and a minimum at 12.331 kPa corresponding to the pressure of pure *n*-heptane. The excess molar Gibbs energy G_m^E of the ternary system (figures 2 and 4) shows a positive deviation from ideality, which is always increa-

TABLE 4. Total pressure p for $\{x(\text{CH}_3)_2\text{CHOH} + (1 - x)\text{CH}_3(\text{CH}_2)_5\text{CH}_3\}$ at $T = 313.15$ K, and at various compositions of the liquid x and vapour phases y

x	y	p/kPa	x	y	p/kPa
0.0000	0.0000	12.345	0.5000	0.5047	21.371
0.0428	0.3295	17.974	0.5500	0.5132	21.361
0.1009	0.3975	19.541	0.5504	0.5133	21.355
0.1500	0.4202	20.181	0.6001	0.5236	21.313
0.1996	0.4387	20.595	0.6001	0.5237	21.308
0.2507	0.4557	20.881	0.6494	0.5363	21.217
0.2983	0.4691	21.072	0.6990	0.5517	21.058
0.3480	0.4804	21.204	0.7492	0.5707	20.801
0.4005	0.4898	21.298	0.7995	0.5952	20.390
0.4009	0.4898	21.300	0.8504	0.6303	19.719
0.4498	0.4972	21.350	0.8998	0.6852	18.660
0.4499	0.4972	21.349	0.9504	0.7887	16.895
0.4985	0.5045	21.367	1.0000	1.0000	13.903

sing in the direction of the less ideal binary $\{x(\text{CH}_3)_2\text{CHOH} + (1 - x)\text{CH}_3(\text{CH}_2)_5\text{CH}_3\}$, with a maximum value of $1268 \text{ J} \cdot \text{mol}^{-1}$.

For the binary systems studied, the three-parameter Margules equation leads to the best results with a root-mean-square deviation pressure of 6 Pa for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)\text{CH}_3(\text{CH}_2)_5\text{CH}_3]$ and a maximum deviation of 14 Pa. The four-parameter Margules equation results in the best fit for $[x\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x)(\text{CH}_3)_2\text{CHOH}]$, with a r.m.s.d. pressure of 9 Pa and $\max|\Delta p| = 21$ Pa. And finally, the best representation for $\{x(\text{CH}_3)_2\text{CHOH} + (1 - x)\text{CH}_3(\text{CH}_2)_5\text{CH}_3\}$ is obtained with the six-parameter Margules equation, with a r.m.s.d. pressure of 28 Pa and $\max|\Delta p| = 71$ Pa. All the binary systems show a positive deviation from ideality, which is more pronounced for systems containing isopropanol. The maximum values of G_m^E for the binary systems have the values of 100, 771 and $1268 \text{ J} \cdot \text{mol}^{-1}$, respectively, all of them being close to the values of the equimolar composition of the liquid phase.

Both binary systems containing isopropanol, exhibit azeotropy. The compositions of the azeotropes as determined by the four and six parameter Margules equation, respectively are: $x(\text{azeotrope}) = 0.8856$ and $p(\text{azeotrope}) = 38.071 \text{ kPa}$ for $[x\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x)(\text{CH}_3)_2\text{CHOH}]$, $x(\text{azeotrope}) = 0.5056$ and $p(\text{azeotrope}) = 21.377 \text{ kPa}$ for $\{x(\text{CH}_3)_2\text{CHOH} + (1 - x)\text{CH}_3(\text{CH}_2)_5\text{CH}_3\}$. We have not found any literature values with which to compare our results under the same conditions. The search for ternary azeotropes has been negative over the entire composition range.

TABLE 5. Total pressure p for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K, and at various compositions of the liquid x_1, x_2 and the vapour phases y_1, y_2

x_1	x_2	y_1	y_2	p/kPa
1.0000	0.0000	1.0000	0.0000	37.134
0.6998	0.3002	0.8659	0.1341	30.280
0.6849	0.2938	0.8233	0.1303	31.050
0.6680	0.2866	0.7859	0.1273	31.670
0.6283	0.2695	0.7262	0.1237	32.366
0.5947	0.2551	0.6932	0.1226	32.535
0.5593	0.2399	0.6673	0.1224	32.516
0.5243	0.2249	0.6470	0.1227	32.378
0.4892	0.2098	0.6295	0.1233	32.158
0.4544	0.1949	0.6135	0.1239	31.871
0.4198	0.1800	0.5981	0.1244	31.524
0.3851	0.1651	0.5825	0.1249	31.112
0.3496	0.1499	0.5658	0.1251	30.621
0.0000	1.0000	0.0000	1.0000	12.351
0.3002	0.6998	0.5733	0.4267	20.693
0.2936	0.6846	0.5045	0.3896	22.642
0.2853	0.6652	0.4542	0.3644	24.016
0.2708	0.6312	0.4071	0.3433	25.070
0.2553	0.5951	0.3796	0.3328	25.542
0.2402	0.5599	0.3614	0.3270	25.749
0.2251	0.5248	0.3467	0.3232	25.828
0.2101	0.4897	0.3337	0.3206	25.823
0.1951	0.4547	0.3213	0.3188	25.754
0.1801	0.4197	0.3090	0.3174	25.639
0.1651	0.3849	0.2966	0.3160	25.478
0.1501	0.3497	0.2838	0.3145	25.270
1.0000	0.0000	1.0000	0.0000	37.126
0.7011	0.0000	0.8037	0.0000	37.165
0.6837	0.0249	0.7876	0.0137	36.610
0.6651	0.0514	0.7704	0.0281	36.043
0.6309	0.1002	0.7388	0.0544	35.031
0.5953	0.1511	0.7061	0.0812	34.005
0.5606	0.2006	0.6746	0.1068	33.029

TABLE 5—*continued*

x_1	x_2	y_1	y_2	p/kPa
0.5255	0.2508	0.6429	0.1325	32.068
0.4908	0.3003	0.6116	0.1578	31.135
0.4546	0.3519	0.5789	0.1844	30.182
0.4206	0.4004	0.5480	0.2099	29.294
0.3855	0.4505	0.5154	0.2370	28.381
0.0000	0.0000	0.0000	0.0000	13.903
0.3001	0.0000	0.6484	0.0000	30.635
0.2929	0.0240	0.6185	0.0292	30.308
0.2859	0.0475	0.5906	0.0564	29.992
0.2705	0.0992	0.5348	0.1104	29.289
0.2552	0.1503	0.4869	0.1557	28.612
0.2409	0.1983	0.4472	0.1924	28.004
0.2252	0.2505	0.4085	0.2273	27.369
0.2103	0.3003	0.3747	0.2574	26.798
0.1953	0.3503	0.3430	0.2851	26.244
0.1802	0.4005	0.3130	0.3113	25.711
0.1653	0.4504	0.2846	0.3360	25.197
0.1653	0.4504	0.2846	0.3360	25.197
0.0000	1.0000	0.0000	1.0000	12.362
0.0000	0.7032	0.0000	0.5315	21.067
0.0331	0.6800	0.0614	0.4965	21.774
0.0466	0.6705	0.0854	0.4827	22.058
0.1013	0.6320	0.1759	0.4298	23.200
0.1484	0.5989	0.2454	0.3885	24.160
0.2000	0.5626	0.3141	0.3477	25.173
0.2498	0.5276	0.3740	0.3122	26.130
0.2997	0.4925	0.4291	0.2799	27.059
0.3500	0.4571	0.4806	0.2502	27.972
0.4004	0.4217	0.5292	0.2228	28.857
0.4498	0.3869	0.5745	0.1978	29.710
0.4998	0.3518	0.6182	0.1744	30.536
0.0000	0.0000	0.0000	0.0000	13.902
0.0000	0.3100	0.0000	0.4515	21.090
0.0290	0.3010	0.0722	0.4101	22.015
0.0471	0.2954	0.1138	0.3866	22.578
0.1080	0.2766	0.2367	0.3181	24.381
0.1485	0.2640	0.3054	0.2804	25.501

TABLE 5—*continued*

x_1	x_2	y_1	y_2	p/kPa
0.1983	0.2486	0.3778	0.2411	26.779
0.2498	0.2327	0.4415	0.2067	28.019
0.2995	0.2173	0.4945	0.1784	29.137
0.3497	0.2017	0.5413	0.1537	30.185
0.3997	0.1862	0.5827	0.1323	31.154
0.4499	0.1707	0.6202	0.1133	32.066
0.4998	0.1552	0.6543	0.0967	32.909

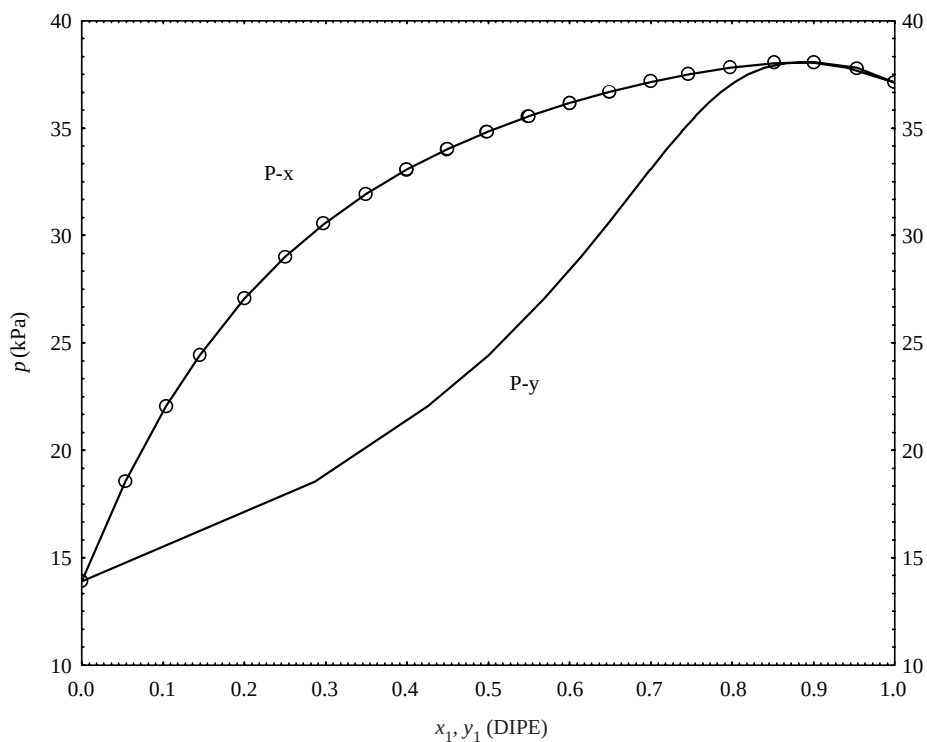


FIGURE 5. Pressure p against liquid composition x and vapour composition y using the four-parameter Margules equation for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15 \text{ K}$: $\circ$, experimental data; —, Margules equation.

TABLE 6. Determined parameters of the models used for the binary subsystems of $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K, together with the coordinates of the azeotrope $x_2(\text{azeotrope})$, and $p(\text{azeotrope})$. The Δp term is defined as the difference between the experimental and calculated pressure

	Margules	Wilson	NRTL		UNIQUAC
$x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)\text{CH}_3(\text{CH}_2)_5\text{CH}_3$					
A_{12}	0.1524	0.9684	0.1551		1.0082
A_{21}	0.1584	0.8821	0.0019		0.9558
$\lambda_{12} = \lambda_{21}$	0.0079				
α_{12}			0.3		
r.m.s.d. $\Delta p/\text{kPa}$	0.006	0.007	0.007		0.007
max $ \Delta p /\text{kPa}$	0.014	0.015	0.015		0.015
$x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x_1)(\text{CH}_3)_2\text{CHOH}$					
A_{12}	1.0988	0.6387	1.0099	1.0131	0.4780
A_{21}	1.4201	0.3513	0.4575	0.5649	1.3139
λ_{12}	0.1902				
λ_{21}	0.4307				
α_{12}			0.47	0.6031	
r.m.s.d. $\Delta p/\text{kPa}$	0.009	0.029	0.056	0.016	0.098
max $ \Delta p /\text{kPa}$	0.021	0.043	0.086	0.035	0.146
$x_1(\text{azeotrope})$	0.8856	0.8863		0.8865	0.8865
$p(\text{azeotrope})/\text{kPa}$	38.071	38.037		37.937	37.934
$x_1(\text{CH}_3)_2\text{CHOH} + (1 - x_1)\text{CH}_3(\text{CH}_2)_5\text{CH}_3$					
A_{12}	2.9678	0.1218	0.9570	1.4278	1.3006
A_{21}	1.9537	0.3470	1.7899	2.1092	0.3550
λ_{12}	5.4125				
λ_{21}	0.7879				
η_{12}	8.3786				
η_{21}	-0.0694				
α_{12}			0.4	0.5366	
r.m.s.d. $\Delta p/\text{kPa}$	0.028	0.152	0.303	0.126	0.402
max $ \Delta p /\text{kPa}$	0.071	0.489	1.112	0.322	1.442
$x_1(\text{azeotrope})$	0.5056	0.5000		0.5056	0.4935
$p(\text{azeotrope})/\text{kPa}$	21.377	21.385		21.320	21.425

TABLE 7. Determined parameters of the models used for $[x_1\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + x_2\text{CH}_3(\text{CH}_2)_5\text{CH}_3 + (1 - x_1 - x_2)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K. The Δp is defined as the difference between the experimental and calculated pressure

	Wohl	Wilson	NRTL	UNIQUAC
C_0	2.3574			
C_1	1.5611			
C_2	1.0714			
A_{12}		0.8362	0.1068	0.8579
A_{21}		1.0119	0.0486	1.1095
A_{13}		0.6570	1.0093	0.4479
A_{31}		0.3371	0.5776	1.3675
A_{23}		0.3764	2.2542	0.3046
A_{32}		0.1050	1.4277	1.4062
α_{12}			0.3	
α_{13}			0.6031	
α_{23}			0.5366	
r.m.s.d. (Δp)/kPa	0.036	0.036	0.025	0.090
max $ \Delta p $ /kPa	0.148	0.121	0.077	0.307

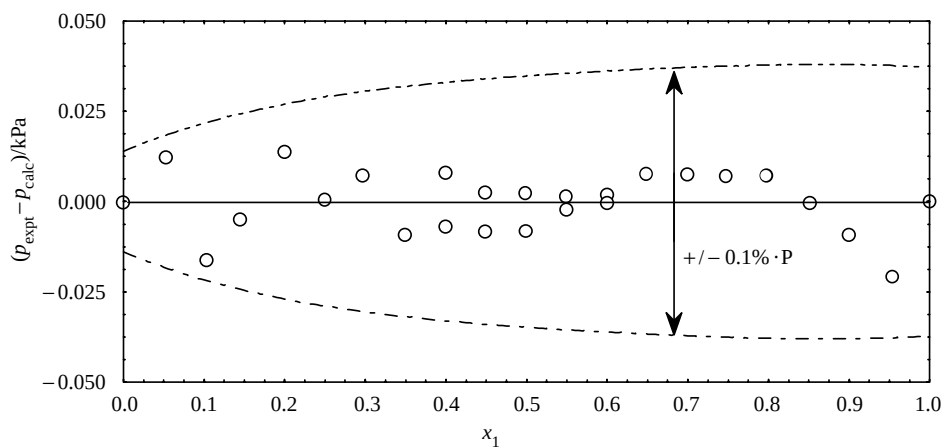


FIGURE 6. Pressure residuals, defined as differences between experimental and calculated pressures using the four-parameter Margules equation, as a function of the DIPE liquid phase composition x for $[x\{(\text{CH}_3)_2\text{CH}\}_2\text{O} + (1 - x)(\text{CH}_3)_2\text{CHOH}]$ at $T = 313.15$ K. The dashed lines indicate ± 0.1 per cent deviation values.

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